

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)



PRESENTS



INNOVATION | DEVELOPMENT | EXECUTION | ADVANCEMENT

Ai - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME (AS SUBMITTED ON PORTAL)		Syntax Weavers	
PROBLEM STATEMENT TITLE		Using Generative AI for Attack Prediction and Remediation	
	NAME	GENDER	AREA OF EXPERTISE
Member 1	Nav Srijan	Male	Machine Learning, Cyber Security
Member 2	Naman Singh	Male	Artificial Intelligence, Flutter
Member 3	Suryansh Gupta	Male	Frontend, React
Member 4	Nikhil Singh	Male	Java, Python, Backend

IDEA TITLE

The "AgamBhitt" Framework: An Agentic Defense Ecosystem

The Concept

Building an autonomous "Overseer" functioning as a high-fidelity Digital Twin of a bank's security posture. It moves beyond isolated tools to create a unified, reasoning defense layer.

The Edge (The Wall)

An inline, Transformer-based WAF that understands semantic intent of HTTP traffic. It blocks advanced evasion techniques (polymorphic payloads) that traditional regex-based WAFs miss.

The Intelligence (The Eyes)

A **Graph Neural Network (GNN)** mapping nodes (servers, users, databases) and edges (permissions, traffic, credentials) to identify hidden "Attack Paths" and predict adversary movement.

The Response (The Brain)

- Generate CACAO-compliant playbooks (executable JSON).
- Draft Root Cause Analysis (RCA) reports for auditing.
- Initiate "Self-Healing" via configuration patches.

Addressing the Core Problem

Modern banking breaches are multi-stage lateral movements. By treating the network as a graph, AgamBhitt solves the "Intelligence Gap" that causes 200+ day detection windows in traditional SOCs.

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

Non-Euclidean Threat Intelligence

While traditional systems look at logs as linear lists, AgamBhatt uses **Graph-Augmented Reasoning**. It understands that a "benign" login at Node A is actually a "Critical Threat" if it's part of a path leading to the Core Banking System.

Predictive Shielding Feedback Loop

This is the most innovative part: the GNN predicts a likely attack vector and **automatically re-tunes the WAF's attention mechanism**. The defense actually hardens itself *before* the attack reaches the target.

Agentic Orchestration

Moving away from static "If-This-Then-That" rules. The LLM acts as an agent that can reason through an incident, cross-reference it with the **MITRE ATT&CK** framework, and decide on the best containment strategy on the fly.

Privacy-Centric LLM Deployment

Banks cannot send sensitive PII to external APIs. Designed to run **Llama 3 or Mistral (8B/70B) locally** on-prem, ensuring data sovereignty while maintaining elite reasoning capabilities.

Proactive Red Teaming

The system includes an "Offensive AI" that continuously probes the network in a sandboxed environment. It finds the "weakest link" and reports it to the "Defensive AI," creating a system that learns from its own failures.

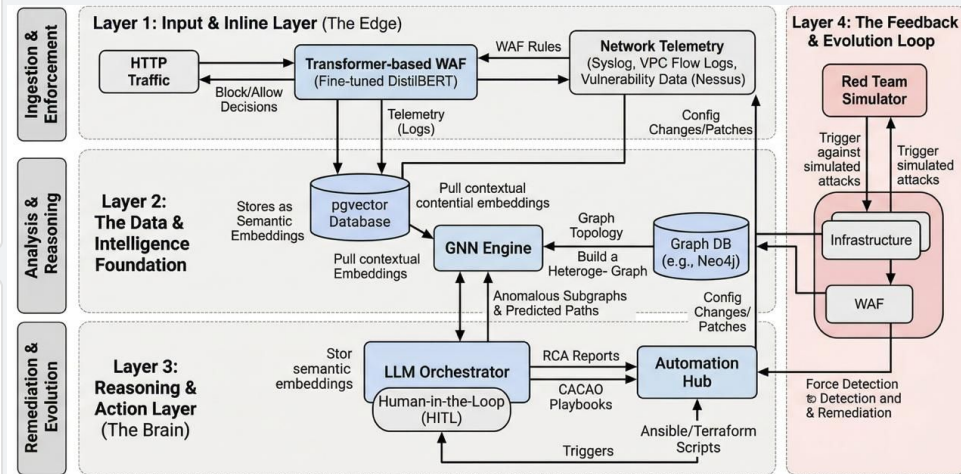
THE STACK & METHODOLOGY

The Tech Stack

- **Backend & API:** FastAPI (for high-concurrency middleware), Python 3.12+
- **AI/ML Models:** PyTorch Geometric (GNN), HuggingFace Transformers (DistilBERT/RoBERTa), Llama 3 / Mistral (hosted on-prem via vLLM)
- **Data Layer:** PostgreSQL with pgvector (for log embeddings), Neo4j (for graph topology)
- **Infrastructure:** Redis (rate-limiting), Docker/Kubernetes (microservices), Ansible (automated patching)

Implementation Methodology

- **Ingestion & Embedding:** Raw logs and network metadata are parsed and converted into semantic vectors using LLM-embeddings
- **Graph Construction:** Entities (IPs, Users, Ports) are mapped as Nodes; interactions are mapped as Edges in a temporal graph
- **Inference Loop:** The GNN identifies anomalous sub-graphs (high-risk paths). The LLM Overseer analyzes these paths to determine intent
- **Enforcement:** The Transformer-WAF enforces blocking at the edge, while the Orchestrator triggers Ansible scripts for internal remediation



FEASIBILITY & VIABILITY

Feasibility Analysis

- **Hardware Ready:** Modern enterprise servers (with NVIDIA A100/H100 GPUs) can handle local LLM inference and GNN training in real-time.
- **Modular Design:** AgamBhatt can be deployed in "Observe Mode" first, allowing banks to see its efficacy without risking downtime.

Potential Challenges & Risks

- **Model Hallucination:** An AI might suggest a remediation that disrupts a critical transaction flow.
- **Infrastructure Latency:** Deep-packet inspection with Transformers can be computationally expensive.

Strategies for Overcoming Challenges

- **Human-in-the-Loop (HITL):** Implementing an **"Approval Gate"** where the AI presents the "Why" and "How," but a human analyst clicks "Deploy" for high-impact changes.
- **Asynchronous Processing:** The WAF handles immediate blocking via optimized **ONNX models**, while the "heavy" GNN/LLM reasoning happens in a parallel out-of-band stream to prevent latency.
- **Explainable AI (XAI):** The LLM is forced to cite specific graph nodes and log entries for every decision, making the defense "transparent" rather than a black box.

IMPACT & BENEFITS

Economic Impact

- **Breach Cost Reduction:** A 2026 data breach in banking averages \$6.4 million. AgamBhitt aims to reduce this by identifying breaches in days rather than months.
- **SOC Efficiency:** Automating L1/L2 analyst work (log parsing, playbook drafting) saves thousands of man-hours per year.

Social & Trust Impact

- **Customer Protection:** Secures the savings of millions by providing a "Zero-Trust" shield that adapts to new AI-driven phishing and fraud.
- **Regulatory Peace of Mind:** Automatically satisfies DORA (Digital Operational Resilience Act) and other global mandates by providing real-time resilience reporting.

Technical Impact

- **Zero-Day Resilience:** Moves the industry from "waiting for a patch" to "proactive architectural hardening."



BUSINESS MODEL

Revenue Streams

- **Subscription Model:** Annual licensing based on the number of endpoints/nodes protected.
- **Professional Services:** High-value implementation fees for custom integration into Core Banking Systems (CBS).
- **Cyber-Range-as-a-Service:** Recurring revenue for continuous, AI-driven red teaming and monthly "Gap Analysis" reports.

Commercialization Potential

- **The Target Market:** Global BFSI (Banking, Financial Services, and Insurance) sector, currently a **\$100B+ cybersecurity market.**
- **Scalability:** The containerized architecture (Docker/K8s) allows the system to scale from small FinTech startups to global mega-banks.

Key Steps to Market

1. **Pilot Program:** Partner with a Neo-bank to test the WAF and GNN on a cloud-native infrastructure.
2. **Marketplace Integration:** Listing AgamBhatt on AWS/Azure Marketplaces as a "Security-Hardened" virtual appliance.
3. **Strategic Partnerships:** Collaborating with cyber-insurance providers to offer lower premiums for banks running the AgamBhatt "Overseer."

RESEARCH & REFERENCES

01. GNNs for Attack Path Prediction

Physics-Informed Graph Neural Networks for Attack Path Prediction

Introduces PIGNN architecture for predictive intelligence with 0.93 F1 score. Validates your project's predictive layer. (MDPI, April 2025)

<https://www.mdpi.com/2624-800X/5/2/15>

02. Transformer-Based WAF & Red Teaming

Generative Pre-Trained Transformer-Based Reinforcement Learning

Details GPT and RL for WAF bypass detection. Supports your evolving USP against adversarial payloads. (IEEE, 2024)

<https://www.computer.org/csdl/journal/tq/2024/01/10059237/1LiKRDe7aZa>

03. LLM-Based Cyberdefense & Big Data

LLMs for Cybersecurity in the Big Data Era: A Comprehensive Review

Analyzes 230+ studies on LLMs for unstructured threat intelligence in financial services. (MDPI, Nov 2025)

<https://www.mdpi.com/2078-2489/16/11/957>

04. Autonomous Red Teaming Agents

AutoRedTeamer: Autonomous Red Teaming with Lifelong Attack Integration

Multi-agent architecture for end-to-end red teaming. Foundation for Agentic Attack Simulation. (NeurIPS, 2025)

<https://neurips.cc/virtual/2025/poster/115242>

05. Hybrid GNN + LLM Security Architectures

CyberLLM-FINDS 2025: Instruction-Tuned LLMs with RAG and Graph Integration

Discusses integrating graph reasoning and RAG into LLMs for MITRE ATT&CK evaluation, mirroring AgamBhatt's core strategy. (arXiv, Jan 2026)

<https://arxiv.org/html/2601.06779v1>

ACCESS THE FULL PROJECT OVERVIEW

<https://drive.google.com/file/d/1KHMMy3Evfl165840Ed1bLMbwihCZNnd5P/view?usp=sharing>

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